

Wenbin Wang, Master.

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Research Interest

My research lies at the intersection of machine learning, statistics, and optimization, with a focus on the theoretical foundations of large language models and the development of AI algorithms for large-scale and multi-agent decision-making. I am particularly interested in understanding the mechanisms of large language models, including their training and prompting processes, and leveraging these insights for applications like decision-making. Recently, my work has focused on reinforcement learning from human feedback (RLHF) and the mathematical foundations of reinforcement learning.

Education

09/2023 – 06/2026	📖 M.Eng. Engineering	Computer Science and Technology
	Advisor: Prof. Ziping Zhao & Prof. Xavier Lagorce	ShanghaiTech University
09/2018 – 06/2022	📖 B.Econ. Economics	Economic Statistics
	Advisor: Prof. Zeyu Zhang	Huazhong Agricultural University

Professional Experience

07/2024 – 09/2024	📖 Research Intern (remote)	Duke University
	Worked on federated unlearning.	Advised by Prof. Minghong Fang
04/2023 – 08/2023	📖 Visiting Student	ShanghaiTech University
	Worked on compressed sensing.	Advised by Prof. Ziping Zhao
09/2022 – 03/2023	📖 Research Intern (remote)	KU Leuven University
	Worked on 3D Reconstruction.	Advised by Dr. Xueyang Kang

Research Publications

Journal Articles

- 1 **W. Wang** and Z. Zhao, “Compressive covariance matrix sensing via quadratic sampling based on nonconvex learning”, *Manuscript in preparation for submission to IEEE Trans. Inf. Theory*, 2025.
- 2 **W. Wang** and Z. Zhao, “Large low-rank covariance matrix estimation with oracle rate via majorization-minimization”, *Manuscript in preparation for submission to IEEE Trans. Signal Process.*, 2025.
- 3 **W. Wang***, Q. Ma*, Z. Zhang, Y. Liu, Z. Liu, and M. Fang, “Federated unlearning under attack: Poisoning threats and defensive strategies”, *Submitted to IEEE Trans. Inf. Forensics Security*, 2025, Under review.

Conference Proceedings

- 1 **W. Wang** and Z. Zhao, “Accelerating graph analytics via bilinear sketching”, in *Manuscript*, 2026.
- 2 **W. Wang**, Z. Jing, Y. Shi, and Z. Zhao, “Low-rank covariance matrix estimation from quadratic measurements”, in *Manuscript*, Under review, 2025.
- 3 **W. Wang**, Y. Shi, and Z. Zhao, “High-dimensional tensor regression with oracle properties”, in *Proc. The International Conference on Machine Learning (ICML)*, 2025. [\[Paper\]](#).

- 4 **W. Wang**, J. Zhang, Y. Yan, J. Zeng, and Z. Zhao, "Achieving oracle statistical property for low-rank matrix recovery with application to image denoising", in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, Under review, 2025.
- 5 **W. Wang** and Z. Zhao, "Achieving oracle rate for large covariance matrix estimation from quadratic measurements", in *Proc. IEEE International Symposium on Information Theory (ISIT)*, 2025.
- 6 **W. Wang** and Z. Zhao, "Optimal compressive covariance sketching via rank-one sampling", in *Proc. The Int. Conf. Sampling Theory Appli. (SampTA)*, 2025. [\[Paper\]](#).
- 7 **W. Wang***, Q. Ma*, Z. Zhang, Y. Liu, Z. Liu, and M. Fang, "Poisoning attacks and defenses to federated unlearning", in *Proc. The Web Conference (WWW)*, 2025. [\[Paper\]](#).
- 8 **W. Wang*** and H. Wang*, "Robust constrained reinforcement learning with linear function approximation", in *Manuscript in preparation for submission to Neural Information Processing Systems (NeurIPS)*, 2025.

Skills

Languages	Strong reading, writing and speaking competencies for English, Mandarin Chinese.
Coding	C/C++, Python, R, $\text{\LaTeX}$, Tensorflow, Pytorch, MATLAB.
Developer Tools	Git, Docker, Visual Studio Code, Jupyter Notebook, Linux Shell.
Misc.	Academic research, teaching, training, $\text{\LaTeX}$ typesetting and publishing.

Miscellaneous Experience

Awards and Achievements

2025	IEEE ISIT Travel Grant Award
2022	Outstanding Graduate , College of Economics and Management, Huazhong Agricultural University.
2020	First Prize , Hubei Provincial Division, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM). Practical Innovation Award , College of Economics and Management, Huazhong Agricultural University. Academic Progress Award , College of Economics and Management, Huazhong Agricultural University. Second Prize , 10th MathorCup College Mathematical Modeling Challenge. First Prize , 17th May 1st Mathematical Contest in Modeling. Second Prize , 13th "Certification Cup" National Undergraduate Mathematical Modeling Challenge.
2019	Merit Student Award , Huazhong Agricultural University.

Teaching

Fall 2024	Teaching Assistant . CS182: Introduction to Machine Learning.
Spring 2025	Teaching Assistant . SI231B: Matrix Computations.

Professional Activity

2025	Conference Reviewer . ICASSP2025.
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